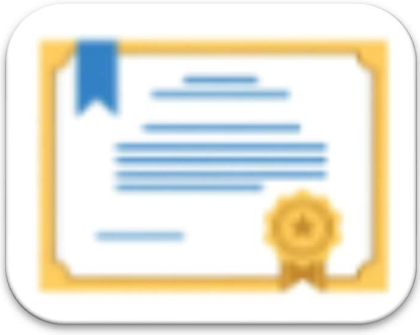
CERTIGEN: AUTOMATED CERTIFICATE GENERATION AND MANAGEMENT SYSTEM

PROJECT DOCUMENTATION

24MNC201



SUBMITTED BY: *Agnes Benoy*

ROLL NUMBER: 12

INSTITUTION: AMAL JYOTHI COLLEGE OF ENGINEERING KOOVAPPALIY

PROJECT TITLE: CertiGen – Automated Certificate Generation and Management System

TOOLS USED: Microsoft Excel, Microsoft Word, Mail Merge, and PDF Export

1. System Description

CertiGen is a certificate generation system developed using Microsoft Excel and Microsoft Word. It uses the Mail Merge feature to generate personalized certificates from student information stored in an Excel worksheet.

The Excel workbook is used to organize student details and assessment results. Microsoft Word is used to design a certificate template containing placeholders for student-specific information. By connecting the Excel data source to the Word document, certificates can be generated for multiple students without manually preparing each certificate.

The system aims to reduce repetitive work, maintain consistent certificate formatting, and minimize errors caused by repeated manual data entry.

1.1 Main Components

Microsoft Excel: Stores student details and assessment information in a structured table.

Microsoft Word: Provides the certificate template and its layout.

Mail Merge: Connects the Excel worksheet to the Word template and inserts the appropriate student details into each certificate.

PDF Export: Can be used to save the generated certificates in PDF format, if included in the final implementation.

1.2 System Requirements

- A computer or laptop.
- Microsoft Excel.
- Microsoft Word with Mail Merge support.
- Student information organized in an Excel worksheet.
- A prepared certificate template.

2. Mail Merge Procedure

The following steps describe the procedure for generating certificates using Mail Merge.

Step 1: Prepare the Excel Data Source

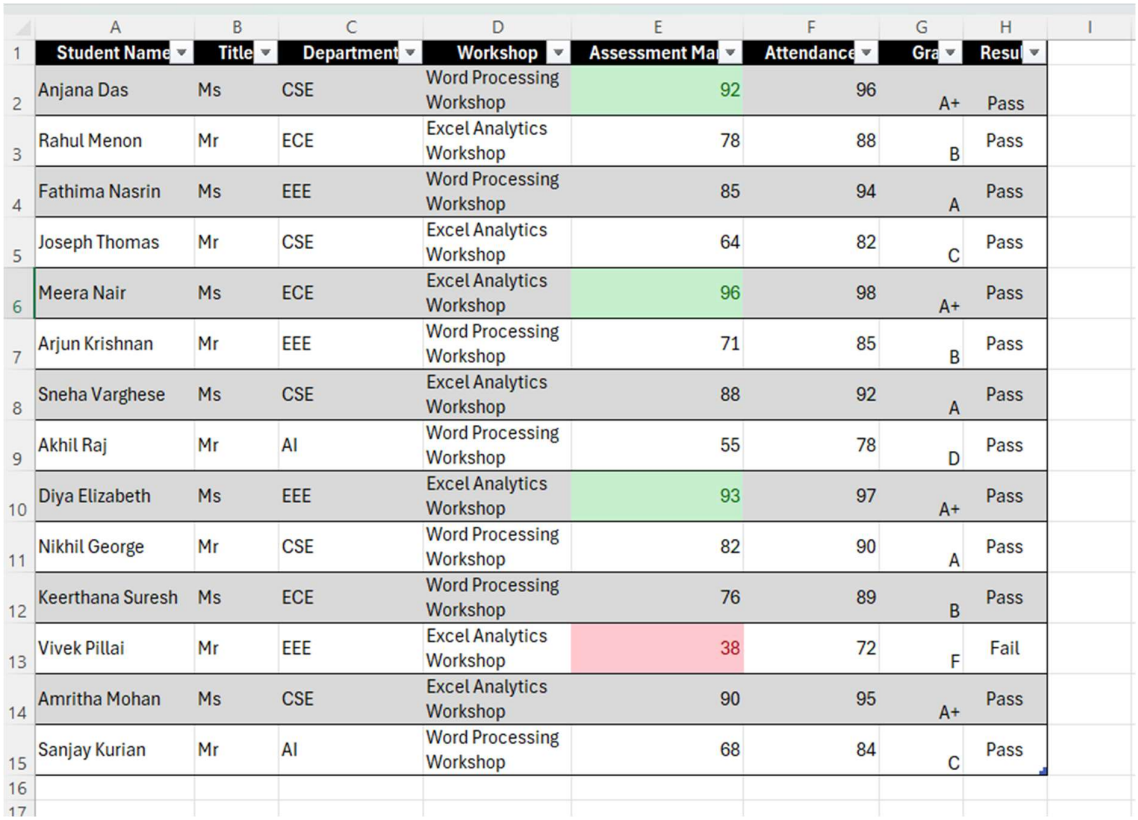
Create an Excel worksheet containing the required student details. Each column should have a clear heading, and each row should represent one student.

Example fields include:

- Student Name
- Department

- Workshop
- Assessment Marks
- Grade
- Result

Screenshot 1: Student Data in Microsoft Excel



	A	B	C	D	E	F	G	H	I
1	Student Name	Title	Department	Workshop	Assessment Ma	Attendance	Gra	Resu	
2	Anjana Das	Ms	CSE	Word Processing Workshop	92	96	A+	Pass	
3	Rahul Menon	Mr	ECE	Excel Analytics Workshop	78	88	B	Pass	
4	Fathima Nasrin	Ms	EEE	Word Processing Workshop	85	94	A	Pass	
5	Joseph Thomas	Mr	CSE	Excel Analytics Workshop	64	82	C	Pass	
6	Meera Nair	Ms	ECE	Excel Analytics Workshop	96	98	A+	Pass	
7	Arjun Krishnan	Mr	EEE	Word Processing Workshop	71	85	B	Pass	
8	Sneha Varghese	Ms	CSE	Excel Analytics Workshop	88	92	A	Pass	
9	Akhil Raj	Mr	AI	Word Processing Workshop	55	78	D	Pass	
10	Diya Elizabeth	Ms	EEE	Excel Analytics Workshop	93	97	A+	Pass	
11	Nikhil George	Mr	CSE	Word Processing Workshop	82	90	A	Pass	
12	Keerthana Suresh	Ms	ECE	Word Processing Workshop	76	89	B	Pass	
13	Vivek Pillai	Mr	EEE	Excel Analytics Workshop	38	72	F	Fail	
14	Amritha Mohan	Ms	CSE	Excel Analytics Workshop	90	95	A+	Pass	
15	Sanjay Kurian	Mr	AI	Word Processing Workshop	68	84	C	Pass	
16									
17									

Figure 1: Student information organized in an Excel worksheet.

Step 2: Design the Certificate Template

Open Microsoft Word and create the certificate layout. Add the certificate title, student details, workshop information, and other relevant text. Leave appropriate spaces for personalized information.

Screenshot 2: Certificate Template in Microsoft Word



Figure 2: Certificate template designed in Microsoft Word.

Step 3: Start Mail Merge

Open the certificate document in Word. Navigate to the **Mailings** tab, select **Start Mail Merge**, and choose the appropriate document type, such as Letters, depending on the template setup.

Step 4: Select the Excel Data Source

Choose **Select Recipients → Use an Existing List**. Browse to the Excel workbook and select the worksheet containing the student records.

Step 5: Insert Merge Fields

Place the cursor where each personalized detail should appear. Use **Insert Merge Field** to insert the corresponding Excel column fields, such as Student_Name, Department, Workshop, Assessment_Marks, and Grade.

Step 6: Preview the Certificates

Select **Preview Results** to check how the student information appears in the certificate. Move through the records to verify that the details are inserted correctly.

Screenshot 3: Mail Merge Fields and Preview Results



Figure 3: Personalized student details displayed using Mail Merge.

Step 7: Filter Records if Required

If certificates are intended only for eligible students, use **Edit Recipient List** to filter the records according to the required condition, such as Result = Pass. Apply this step only if the project uses result-based selection.

Step 8: Complete the Merge

Select **Finish & Merge** and choose the appropriate option to generate the certificates for the selected records.

Screenshot 4: Generated Certificates





Figure 4: Certificates generated from the Excel data source.

Step 9: Save or Export the Output

Save the merged document. If PDF output is required and has been implemented, export the certificates to PDF and verify the final pages.

Screenshot 5: Saved Output or PDF Certificate



Figure 5: Final certificate output.

3. Testing

Testing is performed to check whether the system generates certificates correctly and consistently.

Test Case	Testing Procedure	Expected Result

1. Excel data validation	Check the student records and column headings.	The data is organized correctly.
2. Data source connection	Connect the Excel workbook to Word.	Word recognizes the selected worksheet and records.
3. Merge field insertion	Insert the required fields into the template.	Each field corresponds to the correct column.
4. Certificate preview	Preview certificates for different students.	The correct student details appear in the template.
5. Record selection	Apply a filter if eligibility-based selection is used.	Only the intended records are selected.
6. Multiple certificate generation	Complete the merge for the selected records.	A personalized certificate is generated for each selected record.
7. Formatting verification	Check the layout, alignment, and text placement.	The certificate is readable and consistently formatted.
8. Output verification	Open the saved document or exported PDF.	The output opens correctly and contains the generated certificates.

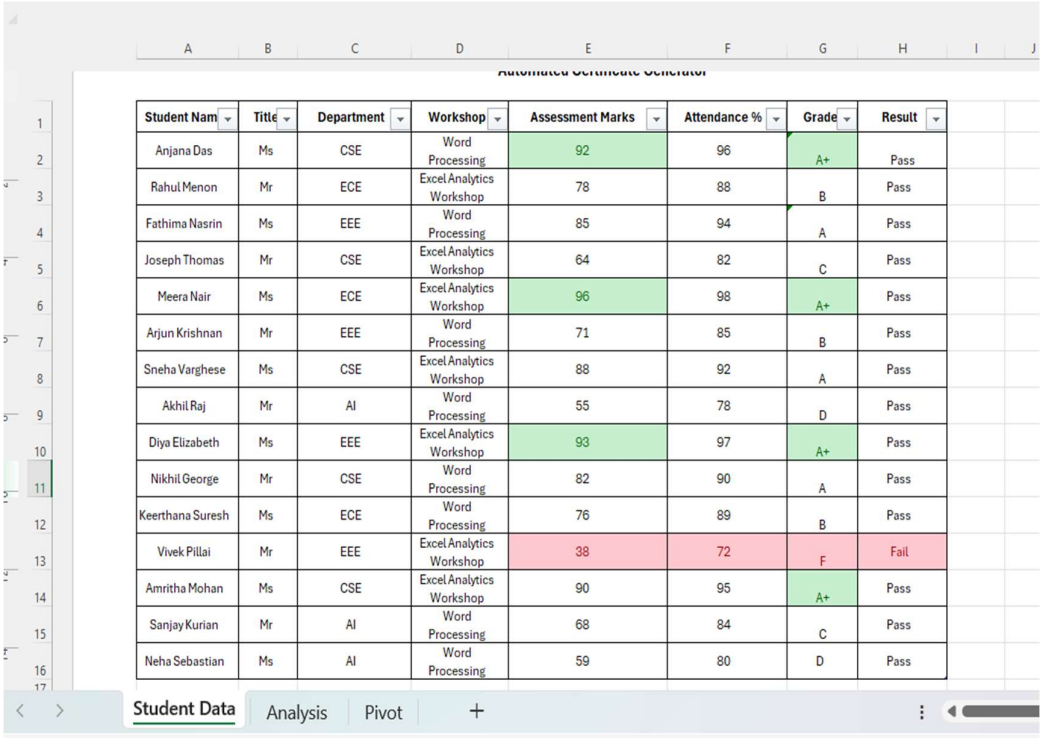
3.1 Testing Results

Complete this section after testing the actual project.

- **Data validation:** [Pass/Fail]
- **Mail Merge connection:** [Pass/Fail]
- **Correct field insertion:** [Pass/Fail]
- **Certificate generation:** [Pass/Fail]
- **Formatting verification:** [Pass/Fail]
- **PDF export, if implemented:** [Pass/Fail/Not Applicable]

4. Screenshots

1. **Excel Student Data:** Shows the student records and column headings.



Student Nam	Title	Department	Workshop	Assessment Marks	Attendance %	Grade	Result
Anjana Das	Ms	CSE	Word Processing	92	96	A+	Pass
Rahul Menon	Mr	ECE	Excel Analytics Workshop	78	88	B	Pass
Fathima Nasrin	Ms	EEE	Word Processing	85	94	A	Pass
Joseph Thomas	Mr	CSE	Excel Analytics Workshop	64	82	C	Pass
Meera Nair	Ms	ECE	Excel Analytics Workshop	96	98	A+	Pass
Arjun Krishnan	Mr	EEE	Word Processing	71	85	B	Pass
Sneha Varghese	Ms	CSE	Excel Analytics Workshop	88	92	A	Pass
Akhil Raj	Mr	AI	Word Processing	55	78	D	Pass
Diya Elizabeth	Ms	EEE	Excel Analytics Workshop	93	97	A+	Pass
Nikhil George	Mr	CSE	Word Processing	82	90	A	Pass
Keerthana Suresh	Ms	ECE	Word Processing	76	89	B	Pass
Vivek Pillai	Mr	EEE	Excel Analytics Workshop	38	72	F	Fail
Amritha Mohan	Ms	CSE	Excel Analytics Workshop	90	95	A+	Pass
Sanjay Kurian	Mr	AI	Word Processing	68	84	C	Pass
Neha Sebastian	Ms	AI	Word Processing	59	80	D	Pass

Student Data | Analysis | Pivot

Figure 6: Excel Student Data.

2. **Excel Analysis:** Shows any calculations or analysis included in the workbook.

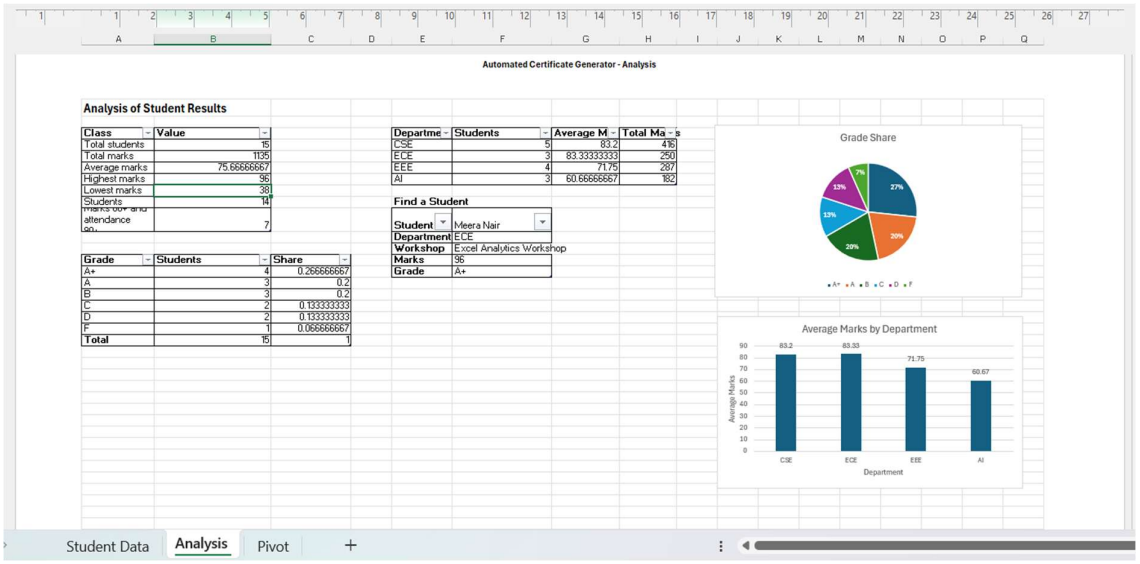


Figure 7: Excel Analysis.

3. **Word Certificate Template:** Shows the certificate design before merging.



Figure 8: Word Certificate Template.

4. **Mail Merge Setup:** Shows the selected data source and merge fields.

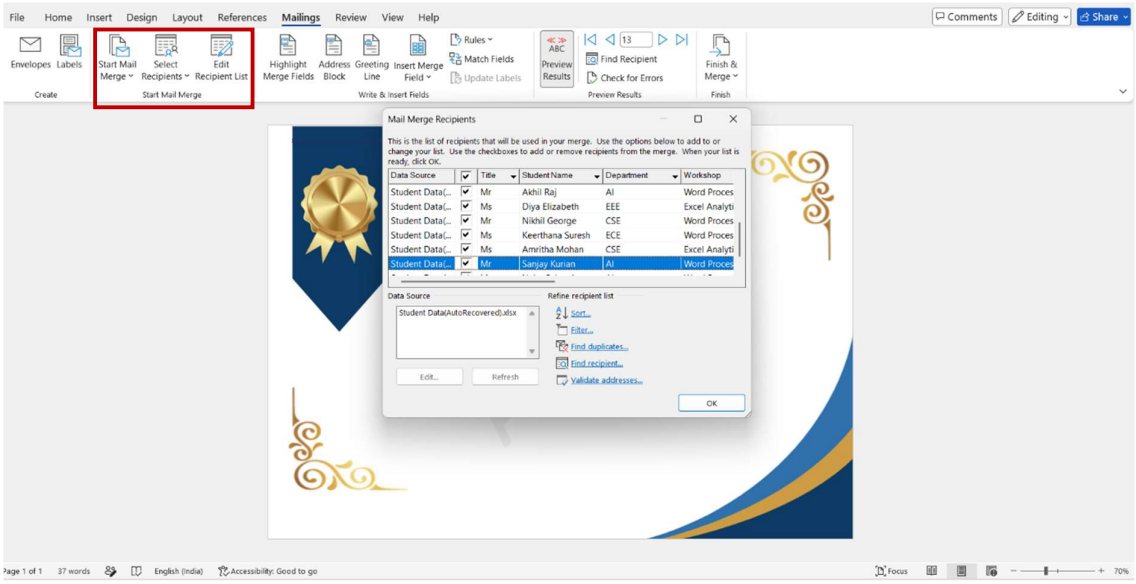


Figure 9: Word Certificate Template

5. **Preview Results:** Shows a certificate populated with student details.

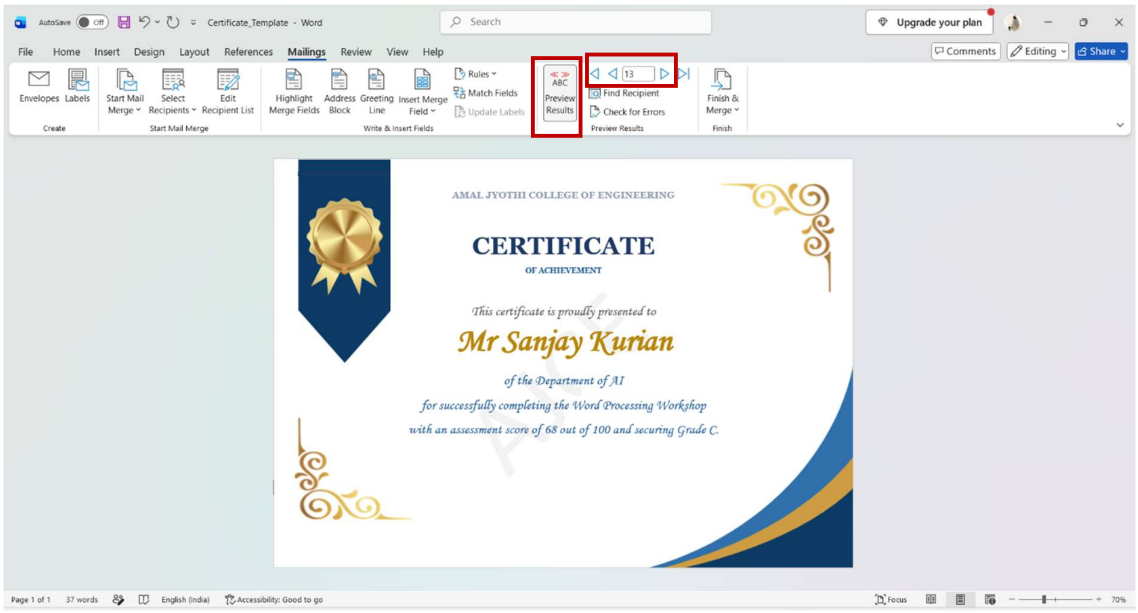


Figure 10: Preview Results

6. **Final Generated Certificates:** Shows the completed merged output.










AMAL JYOTHI COLLEGE OF ENGINEERING
CERTIFICATE
OF ACHIEVEMENT
This certificate is proudly presented to
Ms Meera Nair
of the Department of ECE
for successfully completing the Excel Analytics Workshop
with an assessment score of 96 out of 100 and securing Grade A+.


AMAL JYOTHI COLLEGE OF ENGINEERING
CERTIFICATE
OF ACHIEVEMENT
This certificate is proudly presented to
Mr Arjun Krishnan
of the Department of EEE
for successfully completing the Word Processing Workshop
with an assessment score of 71 out of 100 and securing Grade B.


AMAL JYOTHI COLLEGE OF ENGINEERING
CERTIFICATE
OF ACHIEVEMENT
This certificate is proudly presented to
Ms Sneha Varghese
of the Department of CSE
for successfully completing the Excel Analytics Workshop
with an assessment score of 88 out of 100 and securing Grade A.


AMAL JYOTHI COLLEGE OF ENGINEERING
CERTIFICATE
OF ACHIEVEMENT
This certificate is proudly presented to
Mr Akhil Raj
of the Department of AI
for successfully completing the Word Processing Workshop
with an assessment score of 55 out of 100 and securing Grade D.


AMAL JYOTHI COLLEGE OF ENGINEERING
CERTIFICATE
OF ACHIEVEMENT
This certificate is proudly presented to
Ms Diya Elizabeth
of the Department of EEE
for successfully completing the Excel Analytics Workshop
with an assessment score of 93 out of 100 and securing Grade A+.


AMAL JYOTHI COLLEGE OF ENGINEERING
CERTIFICATE
OF ACHIEVEMENT
This certificate is proudly presented to
Mr Nikhil George
of the Department of CSE
for successfully completing the Word Processing Workshop
with an assessment score of 82 out of 100 and securing Grade A.


AMAL JYOTHI COLLEGE OF ENGINEERING
CERTIFICATE
OF ACHIEVEMENT
This certificate is proudly presented to
Ms Keerthana Suresh
of the Department of ECE
for successfully completing the Word Processing Workshop
with an assessment score of 76 out of 100 and securing Grade B.


AMAL JYOTHI COLLEGE OF ENGINEERING
CERTIFICATE
OF ACHIEVEMENT
This certificate is proudly presented to
Ms Amritha Mohan
of the Department of CSE
for successfully completing the Excel Analytics Workshop
with an assessment score of 90 out of 100 and securing Grade A+.





5. Conclusion

CertiGen demonstrates how Microsoft Excel, Microsoft Word, and Mail Merge can be used together to simplify certificate generation. The system uses structured student data and a reusable certificate template to generate personalized documents efficiently.

The project provides practical experience in spreadsheet organization, document formatting, data integration, and automated document generation. Testing and screenshots help demonstrate whether the implemented system meets its intended requirements.